Chapter 11

The Quantum Formalism

11.1 Introduction

Welcome to Phys 438B! This chapter consolidates the mathematical structure of quantum mechanics in a form suitable for the topics that follow. Rather than revisiting the full axiomatic development from 438A, the goal here is to sharpen and extend the formalism in directions that will be used repeatedly. A key extension introduced is the density operator, which allows a unified treatment of pure states and nonpure states (also called "mixed states"). The density operator is essential when dealing with composite systems, where even a perfectly well-defined global state may induce mixed states locally. The density operator formalism serves as the natural bridge between single-particle quantum mechanics and many-body and even open systems.

We also develop the structure of composite systems using tensor products. This provides the setting for entanglement, correlations, and the addition of angular momentum. The latter will be treated both as a physical example and as a representation-theoretic construction, emphasizing how total angular momentum emerges from combining subsystems. Spin systems, in particular, offer a clean and finite-dimensional context in which these ideas can be worked out explicitly.

Another organizing principle introduced here is the role of generators of transformations. Continuous symmetries, such as translations and rotations, are associated with Hermitian operators that generate unitary transformations. In this framework, operators are not assigned by analogy with classical expressions, but are defined by how they act on states under symmetry transformations. This approach has several advantages: it avoids ambiguities associated with promoting classical expressions to operators; it applies equally well in systems without a clear classical starting point (e.g., spin); it makes symmetry principles primary, clarifying the connection between conservation laws and operators; lastly, it generalizes naturally to more abstract settings, including quantum information and field theory.

The intent of this chapter is not only to review familiar concepts, but to reframe them in a way that supports a more systematic and abstract use of quantum mechanics. The topics that follow rely heavily on this formulation, and benefit from treating it as the primary language rather than as background material.

11.2 Postulates of Quantum Mechanics

The connection between quantum theory and linear algebra was first exposed by Heisenberg's matrix mechanics and later extended by Schrödinger's wave mechanics. A unified theory based on a mathematical structure consisting of *linear operators acting on a vector space* was first formulated by Paul Dirac and later strengthened by John von Neumann, who developed a large portion of functional analysis. Informally, functional analysis is an extension of linear algebraic concepts to infinite-dimensional vector spaces, and while it underlies the formalism of quantum mechanics, we will emphasize only the most essential structures and defer a full treatment to advanced courses.

The postulates of quantum mechanics dictate how to treat a quantum mechanical system mathematically and how to interpret the mathematics to learn about the physical system in question. These postulates cannot be proven, but they have been successfully tested by many experiments, so we accept them as an accurate way to describe quantum mechanical systems. Quantum mechanics has five basic rules, and I've come up with a silly mnemonic to help you remember them:

Small **O**bjects **B**reak **C**lassical **T**heories

11.2.1 The Density Operator

A *quantum experiment* consists of two basic stages: *preparation* and *measurement*. The preparation procedure specifies how the system is produced, while the measurement procedure determines which observable is probed and how outcomes are recorded. In Chapter #1, we considered a Stern–Gerlach experiment in which a beam of thermal silver atoms passes through an analyzer. By blocking the atoms emerging from one output port and allowing only those from the other port to continue, we effectively prepare the system in a definite spin state. Repeating this procedure produces an ensemble of atoms all in the same state, corresponding to a pure spin state consistent with the selected measurement outcome.

Now consider instead the atoms emerging directly from the oven, before any such filtering is applied. The beam contains atoms in a variety of spin states, with no preparation that singles out a definite orientation. Measurements performed on this ensemble yield probabilistic outcomes that cannot be attributed to a single state vector. Rather, the beam must be described statistically, as a collection of systems prepared under identical macroscopic conditions but not in identical quantum pure states.

This distinction motivates a more general description of quantum states. While a state vector suffices for ensembles prepared in a definite pure state, it is not adequate for situations involving incomplete preparation or statistical mixtures. To describe such ensembles, we introduce the **density operator**, which encodes all measurable statistical properties of the system and provides a unified framework for both pure states and mixtures.

As a concrete motivating example, suppose that 50% of the atoms emerging from the oven are found to have $S_z = \hbar/2$ and 50% have $S_z = -\hbar/2$. From this information alone, *we cannot determine the quantum state of the atoms.* In particular, these measurement outcomes are consistent

with very different physical situations. One possibility is that each atom is prepared in the same pure state

$$|\psi\rangle = \tfrac{1}{\sqrt{2}}\Big(|+z\rangle + e^{i\phi}\,|-z\rangle \Big),$$

for some phase ϕ. In this case, the probabilistic outcomes arise from quantum superposition. Alternatively, the beam may consist of a statistical mixture: 50% of the atoms are in the pure state $|+z\rangle$, while the remaining 50% are in the pure state $|-z\rangle$. Informally, the probabilities reflect "classical uncertainty" about which state a given atom occupies. These two situations are fundamentally different, even though they produce identical statistics for measurements of S_z. The difference becomes apparent when we measure a different observable, such as S_x. The coherent superposition exhibits interference effects that depend on the phase ϕ, while the statistical mixture does not.

This example illustrates that specifying probabilities for a single observable is not sufficient to characterize a quantum state. More generally, a complete description must encode the outcomes of all possible measurements. The appropriate object that captures this information, and distinguishes between coherent superpositions and statistical mixtures, is the density operator.

Postulate 1

For every quantum system, there is a complex Hilbert space, and the **states** of the system are all *Hermitian*, *positive*, *trace-class linear maps* ρ for which $\operatorname{Tr}\rho = 1$.

Here are several important properties of the density operator:

(i) Hermitian: $\rho^\dagger = \rho$

(ii) Positive semi-definite: $\langle\psi|\,\rho\,|\psi\rangle \geq 0$ for all $|\psi\rangle$ in the Hilbert space.

(iii) An operator A is **trace-class** if, for any orthonormal basis $\{e_n\}$,

$$\sum_n \langle e_n|\,A\,|e_n\rangle < \infty$$

(iv) Let A be a trace-class operator. Then the **trace** of A is defined as

$$\operatorname{Tr} A = \sum_n \langle e_n|\,A\,|e_n\rangle$$

One can show that the trace does not depend on the choice of orthonormal basis. In a finite-dimensional vector space, the trace of a matrix is the sum of its diagonal entries.

(v) A state is called **pure** if there exists a ket $|\psi\rangle$ such that for all $|\phi\rangle$

$$\rho\,|\phi\rangle = \langle\psi|\phi\rangle\,|\psi\rangle$$

Since a pure state density operator acts as a projection onto a one-dimensional subspace, it can be written in the form

$$\rho = |\psi\rangle \langle\psi| .$$

In this sense, the operator $|\psi\rangle \langle\psi|$ represents the same physical state as the ket $|\psi\rangle$.

This representation is not unique. If $|\psi'\rangle = e^{i\alpha} |\psi\rangle$, then

$$|\psi'\rangle \langle\psi'| = |\psi\rangle \langle\psi| ,$$

so both kets correspond to the same density operator and therefore the same physical state. Thus, pure states are associated not with individual kets, but with one-dimensional subspaces (or equivalently, rays) in Hilbert space.

Example 11.1: Mixed State

Consider a beam of silver atoms emerging directly from an oven and entering a Stern–Gerlach apparatus. Suppose that measurements of S_z yield the outcomes $+\hbar/2$ and $-\hbar/2$ with equal probability. The atoms are produced in a thermal environment and emerge from the oven with *random phases* and no well-defined relative coherence between spin components. Each atom may be viewed as being prepared independently, with no phase relation maintained across the ensemble. This lack of coherence implies that the ensemble cannot be described by a single state vector. Instead, the beam is appropriately described as a statistical mixture, with density operator

$$\rho = \tfrac{1}{2} |+z\rangle \langle+z| + \tfrac{1}{2} |-z\rangle \langle-z| = \tfrac{1}{2} \mathbb{1}.$$

We now phrase the remaining postulates in terms of the density operator ρ. The reader is encouraged to verify that, for a pure state ρ, these reduce to the postulates introduced in Chapter #2. While we will primarily consider pure states, the more general formulation of the quantum state is both more precise and will play an essential role in later chapters.

Postulate 2

Every measurable physical quantity, called an **observable**, is described by a self-adjoint linear operator. The result of measuring an observable is always an eigenvalue of the corresponding operator, and the operator's eigenvectors form a basis for the Hilbert space.

Postulate 3

The probability of obtaining an eigenvalue a_n in the measurement of an observable A on a quantum system in the state ρ is given by the **Born rule**:

$$p(a_n) = \mathrm{Tr}(\rho P_n) \tag{11.1}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n. The expectation value of A may be written as

$$\langle A \rangle = \mathrm{Tr}(\rho A) \tag{11.2}$$

Postulate 4

If a measurement of an observable A on a quantum system in the state ρ yields the result a_n, the quantum system irreversibly **collapses** to a new state

$$\rho' = \frac{P_n \rho P_n}{\mathrm{Tr}(\rho P_n)} \tag{11.3}$$

where P_n is the projection operator of A corresponding to the eigenvalue a_n.

Postulate 5

The **time evolution** of a closed quantum system is described by a *unitary transformation*. That is, the state of the system at time t_0 is related to the state of the system at time t by a unitary operator U which depends on t and t_0 such that

$$\rho(t) = U(t, t_0)\rho(t_0)U^\dagger(t, t_0) \tag{11.4}$$

For continuous evolution, the state obeys the *von Neumann equation* given by

$$i\hbar \frac{d\rho(t)}{dt} = [H(t), \rho(t)] \tag{11.5}$$

where the Hamiltonian $H(t)$ is the observable associated with the total energy of the system and $\hbar$ is the reduced Planck's constant (a universal constant).

11.3 Generators of the Galilei Group

Having established the formal structure of quantum theory, we now apply it to the study of a single *nonrelativistic particle*. By *particle*, we mean a system for which position is, in principle, an observable that can be measured with arbitrary precision. In practice, such localization is limited, and ultimately bounded by relativistic and quantum field–theoretic considerations, which we will not pursue here. By *nonrelativistic*, we mean that the relevant symmetries are those of Galilean transformations relating inertial observers. These symmetries will play a central role, as their generators define the fundamental observables of the theory. Although these symmetries are evidently not fundamental to nature, they provide an accurate and sufficient framework for the systems considered in this course.

If two observers are in inertial frames of reference, their spacetime coordinates are related by the transformation

$$(\mathbf{x}, t) \mapsto (R\mathbf{x} + \mathbf{u}t + \mathbf{a}, t + s)$$

where R denotes a rotation matrix, $\mathbf{u}$ is a velocity vector corresponding to a boost, $\mathbf{a}$ is a displacement vector corresponding to a translation in space, and s is a real number corresponding to a translation in time. These transformations form a continuous group: the Galilei group. Since the laws of nonrelativistic physics are invariant under these transformations, any quantum theory describing such physics must realize this symmetry. This is the core principle underlying our correspondence rules, whereby we identify particular dynamical variables with particular operators.

By Wigner's theorem, any *continuous symmetry transformation* connected to the identity must be induced by a unitary operator acting on the underlying Hilbert space. Galilean transformations are continuous and connected to the identity, so they are represented by unitary operators. Thus, for each Galilean transformation g there is a unitary operator $U(g)$. By Stone's theorem, if $U(g(\epsilon)) = U(\epsilon)$ is an element of a strongly continuous one-parameter unitary subgroup, then there exists a unique self-adjoint operator K such that

$$U(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon K\right), \tag{11.6}$$

where K is called the generator of the transformation, and ϵ is a continuous parameter. For the Galilei group, the generators are:

A **symmetry transformation** is essentially a mapping that preserves all physical predictions. Let U be a unitary operator implementing a continuous symmetry transformation. The predictions of quantum theory are invariant under the following unitary transformation:

$$|\psi\rangle \mapsto U|\psi\rangle$$

$$\rho \mapsto U\rho U^\dagger$$

$$A \mapsto UAU^\dagger$$

Transformation	**Unitary Operator**	**Generator**
Spatial translation ($\mathbf{a}$)	$U(\mathbf{a}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{a}\cdot\mathbf{p}\right)$	p_i (momentum)
Time translation (s)	$U(s) = \exp\left(\frac{i}{\hbar}\,sH\right)$	$-H$ ($-$Hamiltonian*)
Rotation ($R(\boldsymbol{\theta})$)	$U(R) = \exp\left(-\frac{i}{\hbar}\,\boldsymbol{\theta}\cdot\mathbf{J}\right)$	J_i (angular momentum)
Galilean boost ($\mathbf{u}$)	$U(\mathbf{u}) = \exp\left(-\frac{i}{\hbar}\,\mathbf{u}\cdot\mathbf{G}\right)$	G_i (boost generator)

These transformations preserve all probabilities and expectation values:

$$\langle A'\rangle = \langle A\rangle, \quad \mathrm{Tr}(\rho A) = \mathrm{Tr}\big(\rho' A'\big)$$

Our goal is to capture the group structure in the operator representations. Any sequence of spacetime transformations is equivalent to another transformation in the group, so the composition of unitary transformations must respect this property. In fact, the unitary transformation operator U is defined only up to an overall phase, since U and $e^{i\gamma}U$ induce the same transformation on ρ and all observables. Hence, the product of two unitary operators implementing group elements may differ from the operator for the composed group element only by a phase. Generally speaking, this means

$$U(g_1)U(g_2) = e^{i\gamma(g_1,g_2)}U(g_1 \circ g_2) \tag{11.7}$$

for some real-valued function $\gamma(g_1, g_2)$. In many cases, we can determine the phase by additional considerations.

The discussion presented so far is only the starting point of a more detailed analysis, in which the *commutation relations* among the generators are fixed by consistency conditions and transformation properties. In a full treatment[1] , one systematically examines how successive infinitesimal transformations compose, and uses these relations, together with the requirement that the generators transform appropriately under rotations and other symmetries, to determine the "algebra" of the generators. This procedure ultimately leads to the commutation relations of the Galilei group, including the appearance of a central term proportional to the identity. Rather than carry out this derivation in detail, we will simply state the resulting commutation relations, which encode the essential structure:

$$[r_i, p_j] = [G_i, v_j] = i\hbar\delta_{ij}, \quad \text{and} \quad [G_i, p_j] = i\hbar m\delta_{ij}$$

where m is a real-valued constant and v_i is the ith Cartesian component of the **velocity operator**

[1] See Ballentine, 1998.

defined as

$$\mathbf{v} = \frac{i}{\hbar}[H, \mathbf{r}] \tag{11.8}$$

For a free particle with no internal degrees of freedom (i.e., spin), one can derive the following relations between the operators:

$$\mathbf{p} = m\mathbf{v}$$

$$\mathbf{J} = \mathbf{L} = \mathbf{r} \times \mathbf{p}$$

$$H = \tfrac{1}{2}m\mathbf{v}^2 + E_0$$

where $\mathbf{L}$ denotes the orbital angular momentum (a special case of generalized angular momentum), and E_0 is a multiple of the identity. If m denotes the *mass of the particle*, then these expressions are the familiar forms of momentum, angular momentum, and energy from classical mechanics. In the words of Julain Schwinger, we borrow only names from classical mechanics—we've derived these operators from quantum-mechanical principles[2].

For a particle with no internal degrees of freedom interacting with external fields, the most general Hamiltonian consistent with the symmetry requirements is

$$H = \frac{\big[\mathbf{p} - \mathbf{A}(\mathbf{r})\big]^2}{2m} + V(\mathbf{r}) \tag{11.9}$$

where $\mathbf{A}(\mathbf{r})$ is a *vector potential* and $V(\mathbf{r})$ is a *scalar potential*. Vector potentials, as we'll see in the next chapter, are relevant whenever interactions are velocity dependent yet energy conserving. The relation between the momentum and velocity operators is altered by the presence of a vector potential:

$$\mathbf{v} = \frac{\mathbf{p} - \mathbf{A}(\mathbf{r})}{m} \tag{11.10}$$

Observe that adding a constant multiple of the identity to the Hamiltonian does not affect any commutators with observables, and therefore leaves the dynamics unchanged. Hence, the Hamiltonian is defined only up to an additive constant, which we absorb into the definition of $V(\mathbf{r})$. Both $\mathbf{A}(\mathbf{r})$ and $V(\mathbf{r})$ may be time-dependent but must be independent of $\mathbf{p}$.

11.3.1 Spin Degrees of Freedom

In proving the relations summarized thus far in this chapter, we made use of two important mathematical theorems, which we state here without proof:

Theorem. Any operator that commutes with (r_1, r_2, r_3) must be a function of the operators in that set, and likewise for (p_1, p_2, p_3).

[2]Another popular strategy to obtain operators for dynamical variables is called "canonical quantization."

Theorem. For a particle with no internal degrees of freedom, any operator that commutes with $\mathbf{r}$ and $\mathbf{p}$ must be a multiple of the identity.

We now use these results to motivate the introduction of *spin degrees of freedom*. With $\mathbf{J}$ as the generator of rotations (generalized angular momentum) and $\mathbf{L} \equiv \mathbf{r} \times \mathbf{p}$ as the orbital angular momentum, we define the operator

$$\mathbf{S} \equiv \mathbf{J} - \mathbf{L}$$

Using the transformation properties of $\mathbf{r}$ and $\mathbf{p}$ under rotations:

$$[J_i, r_j] = i\hbar\epsilon_{ijk} r_k, \quad [J_i, p_j] = i\hbar\epsilon_{ijk} p_k$$

together with the identities

$$[L_i, r_j] = i\hbar\epsilon_{ijk} r_k, \quad [L_i, p_j] = i\hbar\epsilon_{ijk} p_k$$

it follows immediately that

$$[S_i, r_j] = 0, \quad \text{and} \quad [S_i, p_j] = 0$$

Thus, $\mathbf{S}$ commutes with both $\mathbf{r}$ and $\mathbf{p}$. By Theorem 2, $\mathbf{S}$ must act trivially (i.e., as a multiple of the identity) on a particle with no internal degrees of freedom. However, this is incompatible with with the angular momentum algebra

$$[S_i, S_j] = i\hbar\epsilon_{ijk} S_k$$

unless $\mathbf{S} = 0$. Therefore, for a particle with no internal degrees of freedom, $\mathbf{J} = \mathbf{L}$.

If however, the particle does have internal degrees of freedom, $\mathbf{S}$ may not be proportional to the identity, but it cannot act on the same Hilbert space as $\mathbf{r}$ and $\mathbf{p}$. We introduce an *internal Hilbert space* on which $\mathbf{S}$ acts nontrivially. These internal degrees of freedom are identified with the particle's *spin*. This analysis does not *require* spin; it merely shows that the formalism admits such additional structure. Whether $\mathbf{S} \neq 0$ is ultimately determined by experiment, and empirical observations show that particles carry intrinsic angular momentum independent of their spatial motion.

11.4 Composite Systems

So far we have worked with systems whose degrees of freedom are *kinematically independent*. This includes two important cases: a single particle with multiple types of degrees of freedom (e.g., orbital position/momentum and spin) and multiple commuting observables (e.g., r_1, r_2, and r_3). The defining feature in these examples is that the corresponding operators commute, and the system admits a description in which each degree of freedom can be specified independently. The transition to **composite systems** is conceptually straightforward. We extend this structure to

systems composed of multiple subsystems (e.g., multiple particles). The key requirement is, if subsystems can be prepared and manipulated independently, then the formalism must allow a joint description compatible with their separate descriptions.

Let the operator $A^{(1)}$ represent an observable of component 1, and let $B^{(2)}$ represent an observable of component 2. If $A^{(1)} |a_m\rangle^{(1)} = a_m |a_m\rangle^{(1)}$ and $B^{(2)} |b_n\rangle^{(2)} = b_n |b_n\rangle^{(2)}$, then for every m and n, there must be a joint eigenvector such that

$$A^{(1)} |a_m, b_n\rangle = a_m |a_m, b_n\rangle$$

$$B^{(2)} |a_m, b_n\rangle = b_n |a_m, b_n\rangle$$

These equations are satisfied if the joint eigenvectors are of the product form

$$|a_m, b_n\rangle = |a_n\rangle^{(1)} |b_n\rangle^{(2)} = |a_m\rangle \otimes |b_n\rangle$$

where $\otimes$ denotes the tensor product, and the vectors $\{|a_m\rangle \otimes |b_n\rangle\}$ span a tensor product space. The tensor product between a vector in an M-dimensional space and an N-dimensional space is a vector in an MN-dimensional space.

The **tensor product** of two vector spaces is itself a vector space whose elements are linear combinations of product states denoted $|\psi\rangle \otimes |\phi\rangle$ satisfying bilinearity:

$$\Big(a |\psi_1\rangle + b |\psi_2\rangle\Big) \otimes |\phi\rangle = a |\psi_1\rangle \otimes |\phi\rangle + b |\psi_2\rangle \otimes |\phi\rangle$$
$$|\psi\rangle \otimes \Big(c |\phi_1\rangle + d |\phi_2\rangle\Big) = c |\psi\rangle \otimes |\phi_1\rangle + d |\psi\rangle \otimes |\phi_2\rangle$$

The inner product in a tensor product space is defined multiplicatively as

$$\langle \psi_1 \otimes \phi_1 | \psi_2 \otimes \phi_2 \rangle = \langle \psi_1 | \psi_2 \rangle \langle \phi_1 | \phi_2 \rangle$$

Lastly, for linear operators A on the first space and B on the second, the tensor product $A \otimes B$ is the unique linear operator satisfying

$$(A \otimes B)(|\psi\rangle \otimes |\phi\rangle) = (A |\psi\rangle) \otimes (B |\phi\rangle)$$

Note that there exist vectors in the tensor product space that cannot be written as product states, and likewise, there are linear operators which may not be expressed as a tensor product of two linear operators acting on each space as above.

11.4.1 Statistical Independence

For a two-component system, there exist joint probability distributions that generally do not factorize. This is the basis for entanglement. If, however, the preparations of components 1 and 2 are independent and do not influence each other, then the joint probability distribution for observables $A^{(1)}$ and $B^{(2)}$ should obey the condition of *statistical independence*. If the state of the system is represented by a product state $|\Psi\rangle = |\psi\rangle \otimes |\phi\rangle$, then

$$\text{Prob}(A = a_m \;\&\; B = b_n|\Psi) = |\langle a_m, b_n|\Psi\rangle|^2 = |\langle a_m|\psi\rangle|^2 \, |\langle b_n|\phi\rangle|^2$$

This factorization holds more generally if the state is represented by an operator of the factored form $\rho = \rho^{(1)} \otimes \rho^{(2)}$. It should be emphasized that this factored, or uncorrelated, state is very special and corresponds to independent preparations of the separate components.

11.5 Addition of Angular Momenta

In this section, we present the general formalism corresponding to the problem of adding two commuting angular momenta. Review Chapter #10 for the special case of adding two spin-1/2 particles. At the end of this section, we'll consider some essential examples that will be important for later chapters.

Consider a two-component system, each of which has angular momentum degrees of freedom. For convenience, we'll refer to component 1 as "particle 1" and component 2 as "particle 2," and we denote the corresponding angular momentum operators as $\mathbf{J}_1$ and $\mathbf{J}_2$, although each component may correspond to the orbital and spin degrees of freedom for a single particle (e.g., spin-orbit coupling). Consider the following commuting set of operators:

$$J_1^2 = \mathbf{J}_1 \cdot \mathbf{J}_1 = J_{1x}^2 + J_{1y}^2 + J_{1z}^2$$

$$J_2^2 = \mathbf{J}_2 \cdot \mathbf{J}_2 = J_{2x}^2 + J_{2y}^2 + J_{2z}^2$$

and the corresponding z-components, J_{1z} and J_{2z}. Since these operators commute, they admit a common set of eigenstates. The joint eigenstates are given by tensor products of the individual angular momentum eigenstates:

$$|j_1, m_1\rangle^{(1)} \, |j_2, m_2\rangle^{(2)} = |m_1; m_2\rangle \,,$$

where we drop the labels j_1 and j_2 since they are usually fixed. These eigenstates form the "uncoupled basis," where each factor in the tensor product is an eigenstate of J_i^2 and J_{iz} for $i = 1, 2$. The product space has dimension $(2j_1+1) \times (2j_2+1)$. It is straightforward to verify that the uncoupled basis is complete and orthonormal.

The **total angular momentum operator** is the generator of rotations in the product space,

and it is defined as

$$\mathbf{J} = \mathbf{J}_1 + \mathbf{J}_2 = \mathbf{J}_1 \otimes \mathbb{1} + \mathbb{1} \otimes \mathbf{J}_2 \tag{11.11}$$

It follows that J_1^2, J_2^2, J^2, and J_z mutually commute and therefore admit a common set of eigenstates. In contrast, J_{1z} and J_{2z} cannot be included in this set, since they do not commute with J^2. This non-commutativity arises from the cross term $\mathbf{J}_1 \cdot \mathbf{J}_2$ in J^2, whose components (e.g., $J_{1x}J_{2x}$, $J_{1y}J_{2y}$) do not commute with J_{1z} or J_{2z}.

Designate the joint eigenstates of J_1^2, J_2^2, J^2, and J_z as $|j_1, j_2, j, m\rangle$ such that

$$J_1^2 |j_1, j_2, j, m\rangle = j_1(j_1+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J_2^2 |j_1, j_2, j, m\rangle = j_2(j_2+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J^2 |j_1, j_2, j, m\rangle = j(j+1)\hbar^2 |j_1, j_2, j, m\rangle$$

$$J_z |j_1, j_2, j, m\rangle = m\hbar |j_1, j_2, j, m\rangle$$

Since j_1 and j_2 are usually fixed, we'll use shorthand notation where $|j_1, j_2, j, m\rangle \to |j, m\rangle$. The vectors $|j, m\rangle$ form a complete and orthonormal basis called the "coupled basis," and they span the same product space as the uncoupled basis. The problem of "angular momentum addition" now reduces to finding a transformation between the uncoupled and coupled bases of the product space, i.e., we seek a change of basis from

$$\{|m_1; m_2\rangle\} \to \{|j, m\rangle\}$$

11.5.1 Clebsch-Gordan Coefficients

The uncoupled basis is a *complete basis*, meaning for fixed j_1 and j_2, the uncoupled basis admits the following resolution of the identity on the tensor product space:

$$\sum_{m_1=-j_1}^{j_1} \sum_{m_2=-j_2}^{j_2} |m_1; m_2\rangle \langle m_1; m_2| = \mathbb{1}$$

The coupled basis may be obtained from the uncoupled basis by means of the following transformation, which is obtained by inserting the resolution of the identity above:

$$|j, m\rangle = \Big(\sum_{m_1, m_2} |m_1; m_2\rangle \langle m_1; m_2| \Big) |j, m\rangle$$

$$= \sum_{m_1, m_2} \langle m_1; m_2 | j, m\rangle |m_1; m_2\rangle$$

$$= \sum_{m_1,m_2} C^{j_1 j_2 j}_{m_1 m_2 m} \, |m_1; m_2\rangle$$

The coefficients $C^{j_1 j_2 j}_{m_1 m_2 m} = \langle m_1; m_2 | j, m\rangle$ are called the **Clebsch-Gordan coefficients**, and although they depend on j_1, j_2, j, m_1, m_2, and m, they also obey *selection rules* whereby the coefficients vanish unless certain conditions are satisfied. For example, $C^{j_1 j_2 j}_{m_1 m_2 m}$ vanishes unless $m = m_1 + m_2$. Since $J_z = J_{1z} + J_{2z}$,

$$0 = \langle m_1; m_2 | \left(J_z - J_z^{(1)} - J_z^{(2)} \right) |j, m\rangle$$

$$= \hbar \, (m - m_1 - m_2) \, \langle m_1; m_2 | j, m\rangle$$

which shows that $C^{j_1 j_2 j}_{m_1 m_2 m}$ is nonzero only when $m - m_1 - m_2 = 0$. Evidently, the uncoupled basis states are also eigenvectors of the operator J_z with eigenvalue $(m_1 + m_2)\hbar$.

For a fixed value of m, the eigenvectors of J_z with eigenvalue $m\hbar$ form a subspace (or eigenspace). Both the coupled states $|j, m\rangle$ and uncoupled states $|m_1; m_2\rangle$ with $m_1 + m_2 = m$ are eigenstates of J_z with eigenvalue $m\hbar$. Since the dimension of a subspace (like any vector space) is invariant under a change of basis, the number of coupled states with a given m must equal the number of uncoupled states with $m_1 + m_2 = m$. The eigenspace having the maximum value of $m = j_1 + j_2$ is one dimensional, i.e., there is only one state with $m = j_1 + j_2$. In the uncoupled basis, that state is $|m_1 = j_1; m_2 = j_2\rangle$, where both m_1 and m_2 take their maximum values. In the coupled basis, the state with $m = j_1 + j_2$ must belong to the set of states (called a multiplet) with $j = j_{\max} \equiv j_1 + j_2$. Hence, the uncoupled state $|m_1 = j_1; m_2 = j_2\rangle$ differs only by an overall phase from the coupled state $|j_{\max}, j_{\max}\rangle$.

In the set of states with $j = j_{\max}$, there is a single state with $m = j_{\max} - 1$. However, in the uncoupled basis, the value of $m = m_1 + m_2 = j_{\max} - 1$ occurs *twice*: the corresponding states are $|j_1 - 1; j_2\rangle$ and $|j_1; j_2 - 1\rangle$. Since the $m = j_{\max} - 1$ eigenspace is two dimensional, there must be another coupled state with this value of m. That state must have $j = j_{\max} - 1$, since the $j = j_{\max}$ state has already been accounted for. We can continue this process as long as the allowed ranges of m_1 and m_2 have not been exhausted. At $m = j_{\max} - 2$, the uncoupled basis contains three states, while the $j = j_{\max}$ and $j = j_{\max} - 1$ multiplets each account for one coupled state with this value of m. The remaining state must therefore have $j = j_{\max} - 2$. Continuing downward in this way, each successive value $j = j_1 + j_2, j_1 + j_2 - 1, \ldots$, appears exactly once. The sequence terminates when the total number of states in the coupled basis equals the dimension of the product space.

Since there are $2j+1$ states in the coupled basis for each j, and the product space has dimension $(2j_1 + 1) \times (2j_2 + 1)$, there must exist a unique $j_{\min}$ such that

$$\sum_{j=j_{\min}}^{j_{\max}} (2j + 1) = (2j_1 + 1)(2j_2 + 1).$$

On the left-hand side, we can evaluate the summation explicitly. Let $N = j_{\max} - j_{\min}$ (a non-

negative integer) and re-index the sum by setting $k = j_{\max} - j$ so that k runs from 0 to N as j runs from $j_{\max}$ down to $j_{\min}$. Consequently,

$$\begin{aligned}\sum_{j=j_{\min}}^{j_{\max}} (2j+1) &= \sum_{k=0}^{N} [2j_{\max} + 1 - 2k] \\ &= (2j_{\max}+1)\sum_{k=0}^{N} 1 - 2\sum_{k=0}^{N} k \\ &= (2j_{\max}+1)(N+1) - N(N+1) \\ &= (N+1)(2j_{\max}+1-N)\end{aligned}$$

Equating to the right-hand side and expanding in terms of j_1 and j_2, we find

$$(j_1 + j_2 - j_{\min} + 1)(j_1 + j_2 + j_{\min} + 1) = (2j_1+1)(2j_2+1)$$

or

$$j_{\min}^2 = (j_1 - j_2)^2 \quad \Rightarrow \quad j_{\min} = |j_1 - j_2|$$

The Clebsch-Gordan coefficients vanish unless m and j satisfy the following **selection rules**:

$$\boxed{\begin{aligned} &m = m_1 + m_2, \\ &j = |j_1 - j_2|,\ |j_1 - j_2| + 1,\ \ldots,\ j_1 + j_2. \end{aligned}} \tag{11.12}$$

The Clebsch-Gordan coefficients are taken to be real by convention, and furthermore, for each allowed value of j, the coupled basis states $|j, m\rangle$ are defined such that the Clebsch–Gordan coefficient for the "highest-weight" state satisfies

$$\langle m_1 = j_1\,;\, m_2 = j - j_1 | j, j\rangle > 0, \tag{11.13}$$

i.e., it is real and strictly positive. All other Clebsch–Gordan coefficients are then determined by the action of the ladder operators and by orthonormality[3]. While the Clebsch-Gordan coefficients are tabulated in many books, we'll briefly summarize a procedure to generate them before looking at a few important examples in more detail.

The Clebsch–Gordan coefficients for the highest- and lowest-weight states of the $j = j_1 + j_2$ multiplet are equal to one:

$$\langle m_1 = j_1; m_2 = j_2 | j_1 + j_2, j_1 + j_2\rangle = \langle m_1 = -j_1; m_2 = -j_2 | j_1 + j_2, -(j_1 + j_2)\rangle = 1. \tag{11.14}$$

[3]The phase convention mentioned above is chosen to be consistent with the phase convention used in defining the action of the ladder operators in Chapter #4.

The other coefficients can be related to each other by recursion relations, which can be derived by evaluating the matrix elements of $J_\pm = J_{1\pm} + J_{2\pm}$. From,

$$J_\pm |j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\,|j,m\pm 1\rangle$$

$$J_{1\pm} |m_1;m_2\rangle = \hbar\sqrt{j_1(j_1+1) - m_1(m_1\pm 1)}\,|m_1\pm 1;m_2\rangle$$

$$J_{2\pm} |m_1;m_2\rangle = \hbar\sqrt{j_2(j_2+1) - m_2(m_2\pm 1)}\,|m_1;m_2\pm 1\rangle$$

we get the following recursion relation:

$$\begin{aligned}\sqrt{j(j+1) - m(m\pm 1)}\, C^{j_1 j_2 j}_{m_1 m_2,\, m\pm 1} = {} & \sqrt{j_1(j_1+1) - m_1(m_1\pm 1)}\, C^{j_1 j_2 j}_{m_1\pm 1,\, m_2,\, m} \\ & + \sqrt{j_2(j_2+1) - m_2(m_2\pm 1)}\, C^{j_1 j_2 j}_{m_1,\, m_2\pm 1,\, m}.\end{aligned} \tag{11.15}$$

By orthonormality of the coupled and uncoupled bases, we can derive the following properties of the Clebsch-Gordan coefficients:

$$\sum_{m_1,m_2} C^{j_1 j_2 j}_{m_1 m_2 m}\, C^{j_1 j_2 j'}_{m_1 m_2 m'} = \delta_{jj'}\,\delta_{mm'} \tag{11.16}$$

$$\sum_{j,m} C^{j_1 j_2 j}_{m_1 m_2 m}\, C^{j_1 j_2 j}_{m_1' m_2' m} = \delta_{m_1 m_1'}\,\delta_{m_2 m_2'} \tag{11.17}$$

$$\sum_{m_1,m_2} \left(C^{j_1 j_2 j}_{m_1 m_2 m}\right)^2 = 1 \tag{11.18}$$

11.5.2 Two Spin-1/2 Particles

Let's revisit a familiar problem to learn how the coefficients work. Let $j_1 = \frac{1}{2}$, $j_2 = \frac{1}{2}$. Then $m_1, m_2 = \pm\frac{1}{2}$ and the uncoupled basis states are

$$\left|\tfrac{1}{2};\tfrac{1}{2}\right\rangle,\ \left|\tfrac{1}{2};-\tfrac{1}{2}\right\rangle,\ \left|-\tfrac{1}{2};\tfrac{1}{2}\right\rangle,\ \left|-\tfrac{1}{2};-\tfrac{1}{2}\right\rangle,$$

which we relabel using $\pm$ to indicate *spin up* and *spin down*:

$$|{+}{+}\rangle,\ |{+}{-}\rangle,\ |{-}{+}\rangle,\ |{-}{-}\rangle.$$

By the selection rules, the allowed values of j are 0 and 1. Starting with $j = 1$, the state with $m = 1$ is given by

$$|1,1\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},\frac{1}{2},1}\,|{+}{+}\rangle \quad\Rightarrow\quad C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},\frac{1}{2},1} = 1$$

and the state with $m = -1$ is

$$|1,-1\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},-\frac{1}{2},-1}\,|--\rangle \quad \Rightarrow \quad C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},-\frac{1}{2},-1} = 1$$

The state with $m = 0$ takes a bit more work:

$$|1,0\rangle = C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},-\frac{1}{2},0}\,|+-\rangle + C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},\frac{1}{2},0}\,|-+\rangle$$

Using the recursion relation with $m_1 = 1/2$ and $m_2 = -1/2$,

$$\sqrt{2}C^{\frac{1}{2}\frac{1}{2}1}_{\frac{1}{2},-\frac{1}{2},0} = 1$$

and with $m_1 = -1/2$ and $m_2 = 1/2$,

$$\sqrt{2}C^{\frac{1}{2}\frac{1}{2}1}_{-\frac{1}{2},\frac{1}{2},0} = 1$$

Hence,

$$|1,0\rangle = \tfrac{1}{\sqrt{2}}\Big(\,|+-\rangle + |-+\rangle\,\Big)$$

Having generated the triplet for $j = 1$, consider

$$|0,0\rangle = C^{\frac{1}{2}\frac{1}{2}0}_{\frac{1}{2},-\frac{1}{2},0}\,|+-\rangle + C^{\frac{1}{2}\frac{1}{2}0}_{-\frac{1}{2},\frac{1}{2},0}\,|-+\rangle$$

Since the first coefficient is positive, and the basis states are orthogonal, we have

$$|0,0\rangle = \tfrac{1}{\sqrt{2}}\Big(\,|+-\rangle - |-+\rangle\,\Big)$$

Example 11.2: Hyperfine Structure of the Hydrogen Ground State

In Chapter 9, we found that the energy eigenvalues of the Hamiltonian for the hydrogen atom are given by $E_n = -13.6\ \text{eV}/n^2$, where n is a positive integer. As it turns out, there are a number of corrections that must be made to our original model, and one of those corrections involves the intrinsic spins of both the electron and the proton. Since the proton has a magnetic moment (proportional to its spin), the proton is a source of a magnetic field. The electron interacts with this magnetic field, producing an interaction energy proportional to the intrinsic spins of both particles.

If we restrict our analysis to the ground state of hydrogen, a state with zero orbital angular momentum, the spin-spin Hamiltonian can be expressed in the simple form

$$H = \frac{2A}{\hbar^2}\mathbf{S}_1 \cdot \mathbf{S}_2$$

where $\mathbf{S}_1$ and $\mathbf{S}_2$ denote the spin angular momenta of the electron and proton respectively, and $A > 0$ is a constant. As it turns out, the eigenstates of this Hamiltonian are precisely the coupled eigenstates of two spin-1/2 particles. From the identity,

$$\mathbf{S}^2 = (\mathbf{S}_1 + \mathbf{S}_2)^2 = \mathbf{S}_1^1 + \mathbf{S}_2^2 + 2\mathbf{S}_1 \cdot \mathbf{S}_2,$$

we find

$$2\mathbf{S}_1 \cdot \mathbf{S}_2 = \mathbf{S}^2 - \mathbf{S}_1^1 - \mathbf{S}_2^2,$$

and since the coupled basis is a simultaneous eigenstate of $\mathbf{S}_1^2$, $\mathbf{S}_2^2$, and $\mathbf{S}^2$ (also S_z^2), it follows that a coupled basis vector is also an eigenvector of $\mathbf{S}_1 \cdot \mathbf{S}_2$. The corresponding eigenvalue equations of the Hamiltonian are

$$H\,|1, \pm 1\rangle = H\,|1, 0\rangle = \frac{A}{2}\,|1, 1\rangle\,, \quad H\,|0, 0\rangle = -\frac{3A}{2}\,|0, 0\rangle$$

The triplet states are therefore degenerate with energy $A/2$ and the singlet state has energy $-3A/2$. The energy difference is given by

$$\Delta E = \left| \frac{A}{2} - \left(-\frac{3A}{2} \right) \right| = 2A$$

A photon emitted or absorbed in making a transition between these two energy levels must therefore have a wavelength λ determined by $hc/\lambda = 2A$. For hydrogen, this wavelength can be determined by spectroscopic methods and happens to be 21 cm.

11.5.3 Addition of Orbital and Spin Angular Momentum

An important application of angular momentum addition in atomic physics is the coupling of the orbital angular momentum $\mathbf{L}$ and spin angular momentum $\mathbf{S}$ of electrons. The total angular momentum is defined as

$$\mathbf{J} = \mathbf{L} + \mathbf{S}.$$

For a single electron (e.g., hydrogen), the spin is $s = \frac{1}{2}$. The allowed values of the total angular momentum j are

$$j = \ell + s,\, \ell + s - 1,\, \ldots,\, |\ell - s|.$$

Thus,

$$j = \begin{cases} \ell + \frac{1}{2},\ \ell - \frac{1}{2}, & \ell \geq 1, \\ \frac{1}{2}, & \ell = 0. \end{cases}$$

It is conventional to use lower-case letters (j, ℓ, s) for a single electron and upper-case letters (J, L, S) for the total angular momenta of the atom.

For atoms with multiple electrons, one typically first adds all orbital angular momenta to obtain

L, then all spins to obtain **S**, and finally couples **L** and **S** to obtain **J** (LS coupling). Atomic states are labeled using spectroscopic notation

$$^{2S+1}L_J$$

where S and J are quantum numbers and L is denoted by a letter:

L	0	1	2	3	4	$\cdots$
	S	P	D	F	G	$\cdots$